

B.3 Comparison to Related Work Detectors

Linear Separability and Representation Quality Analyses Compared to prior work on MGT detection based on latent representations (Tulchinskii et al., 2023; Yu et al., 2024; Chen et al., 2025), our study differs both in its analysis of latent representations and in its methodological approach. Tulchinskii et al. (2023) were among the first to investigate latent-space differences between human- and machine-written text. While they show that intrinsic dimensionality differs between the two, we additionally analyze entropy, effective rank, and anisotropy, examine their evolution across layers, and demonstrate that all four metrics exhibit consistent patterns across diverse domains. TextFluoroscropy (Yu et al., 2024) provides evidence that middle layers contain particularly robust MGT signals, but does not investigate the linear separability of human and machine representations. RepreGuard (Chen et al., 2025) identifies systematic differences in activation patterns, but neither tests for linear separability nor connects its findings to the Linear Representation Hypothesis (LRH) (Park et al., 2023).

Methodology Methodologically, our approach differs in several ways. ID (Tulchinskii et al., 2023) trains a logistic regression classifier on a single intrinsic-dimensionality feature per sample, whereas our probes operate directly on latent representations. TextFluoroscropy (Yu et al., 2024) trains a *nonlinear* classifier on the middle-layer representation selected through vocabulary-space distributional differences between early and late layers. In contrast, we train simple *linear* classifiers and show that they achieve stronger performance. RepreGuard (Chen et al., 2025) identifies probing directions through an unsupervised, *training-free* procedure. Specifically, it computes the first principal component of the activation-difference matrix between human and machine text using mean-pooled token representations.²¹ In contrast, we *train* regularized linear classifiers directly on representations and show that last-token pooling consistently outperforms mean pooling (Appendix D). Moreover, RepreGuard operates in the full activation space (e.g., 4,096 dimensions for LLaMA-8B), whereas we first project activations into a 100-dimensional PCA subspace. As our representation analysis shows that MGT and HWT remain sep-

arable in low-dimensional spaces, our probes operate on only 2.4% of the original dimensionality, substantially reducing memory and computational requirements.

Tables 1 and 2 show that MGT probes consistently outperform these related methods. Compared to RepreGuard (Chen et al., 2025), we attribute these gains primarily to supervised, regularized learning, which suppresses noisy dimensions and yields more robust probing directions. Compared to TextFluoroscropy (Yu et al., 2024), our probes avoid unnecessary nonlinear complexity and leverage information from all layers rather than relying on a single middle layer. As shown in Section 3, the boundary between human- and machine-written text is largely linear, while Appendix Figure 15 demonstrates that aggregating information across layers improves robustness. Finally, unlike ID (Tulchinskii et al., 2023), our probes operate directly on latent representations rather than a single-metric feature, enabling them to exploit substantially richer information about machine-generated text.

B.4 Computational Resources

We run all experiments on an HPC cluster equipped with NVIDIA A100 (40GB) and H200 (141GB) GPUs. Training a single probe variant, including activation extraction and inference, typically requires only 1–3 minutes on an A100 (40GB).

C Additional Results

C.1 ID Adversarial Attacks

In addition to the dimensions considered in Table 1 (domains, languages, generators, and generation tasks), we also evaluate adversarial attacks using the DetectRL benchmark (Wu et al., 2024). We observe the same saturation effects as for the DetectRL domain subsets, with most baselines achieving near-perfect performance. For this reason, we omit adversarial attacks from the main table.

C.2 OOD

In Table 2, we report the mean AUC for each target subset unseen during training, averaged over transfers from the remaining subsets within the same benchmark. Figure 10 provides the full OOD transfer results.

²¹This is closely related and often identical to difference-in-means (Tigges et al., 2024; Rimskey et al., 2024).

DetectRL Attacks (Wu et al., 2024)				
Model	Mixing	Paraphrase	Perturbation	Prompt
Zero-shot				
Likelihood	0.9562	0.6973	0.6535	0.9628
LLR	0.9232	0.6813	0.6876	0.9288
Rank	0.2121	0.4113	0.8194	0.1584
Binoculars	0.9992	0.9069	0.9248	0.9847
FastDetectGPT	0.8911	0.7858	0.6218	0.8926
GECsScore	0.7183	0.6777	0.6075	0.7064
RAIDAR	0.7392	0.7209	0.6503	0.7122
Supervised				
OpenAI-RoBERTa	0.8774	0.8991	0.6946	0.8581
RADAR	0.9733	0.9628	0.9593	0.9474
EditLens	0.3302	0.5628	0.5775	0.3562
ID	0.7089	0.7253	0.6406	0.6746
RepreGuard	0.9600	0.8480	0.9820	0.9260
BiScope	0.9993	0.9692	0.9884	0.9946
TextFluoroscropy	0.9944	0.9968	0.9971	0.9934
RoBERTa	0.9964	0.9992	1.0000	0.9972
MGT Probes				
LLP	0.9993	0.9999	1.0000	0.9982
Δ vs BL	+0.00	+0.07	+0.00	+0.10
CLP	0.9995	0.9998	1.0000	0.9994
Δ vs BL	+0.02	+0.06	+0.00	+0.22

Table 3: In-domain detection AUC scores for adversarial attacks on DetectRL (Wu et al., 2024). LLP = Layer-Averaged Linear Probes. CLP=Concatenated-Layer Probes. For TSM, FP=First Paragraph, PE=Paragraph Extension, SUM=Summarization, and TST=Text Style Transfer. Δ denotes the gain over the strongest baseline.

C.3 Probing vector similarity

Figure 11 shows layer-16 activation projections for paired domain subsets in DetectRL (Wu et al., 2024). The black dashed line connects the centroids of the human and machine activation clouds within each subset, approximating the corresponding linear MGT direction.

Consistent with Figure 4, we find that these linear MGT directions are broadly aligned in the same direction across most subsets. They are not perfectly aligned, which is expected because (1) we approximate the MGT direction by connecting class centroids rather than learning it directly, and (2) the subsets differ along several confounding dimensions, including adversarial attacks, text length, and stylistic or grammatical variation. The latter is particularly evident for Reddit, which exhibits the largest distributional shift across domains. Nevertheless, their broad alignment provides further evidence that probes learn an approximate shared linear direction associated with MGT.

C.4 Sample-efficiency Analysis

In Figure 5, we evaluate sample efficiency using a PCA space constructed from the full training set for each layer, and then train probes on progressively

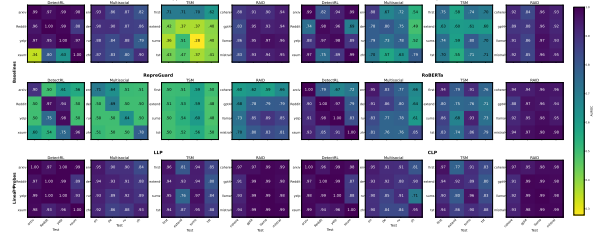


Figure 10: OOD AUC scores for baseline detectors (top rows) and MGT probes (bottom row). Rows denote the training subsets, while columns denote the test subsets. Darker colors indicate better performance.

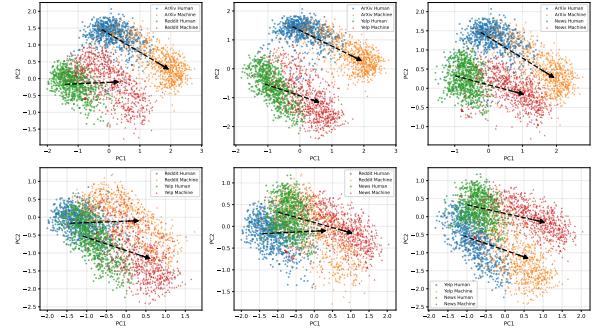


Figure 11: Activation projections of paired DetectRL datasets (Wu et al., 2024) in the shared PCA space at layer 16. Dashed lines connect the centroids of the human and machine clusters within each dataset.

smaller subsets. However, this setup does not fully reflect a real-world application, where the full training distribution may not be available in advance. Therefore, in Figure 12, we repeat the experiment in the full activation space, without applying PCA. We observe the same pattern, confirming that the sample-efficiency results.

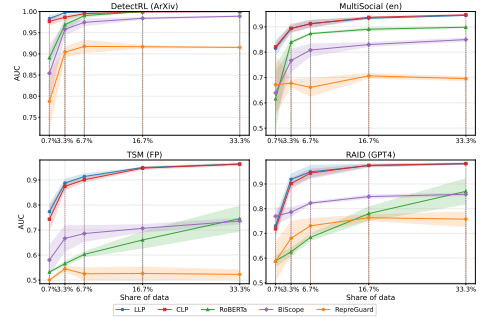


Figure 12: In-domain AUC scores for selected datasets as a function of training set size. Compared to Figure 5, we do not reduce the dimensionality of activations.

C.5 Detecting AI-edited text

In Section 5.6, we investigate whether probing vectors encode MGT signals as a continuum rather

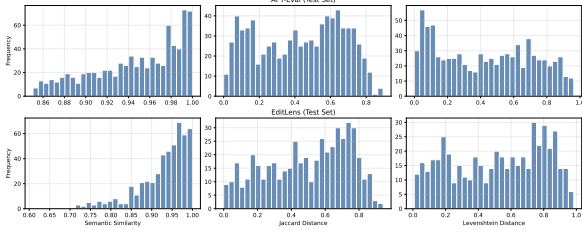


Figure 13: Histograms of the similarity measures Semantic (Cosine) Similarity, Jaccard Distance, and Levenshtein Distance for the test sets of APT (Saha and Feizi, 2025) and EditLens (Thai et al., 2026).

than a binary distinction. Figure 13 shows the distributions of the text-similarity metrics used to quantify the degree of AI editing. Our random sampling procedure yields a relatively uniform distribution across metrics and benchmarks, with the exception of semantic similarity in EditLens, which exhibits a left-skewed distribution.

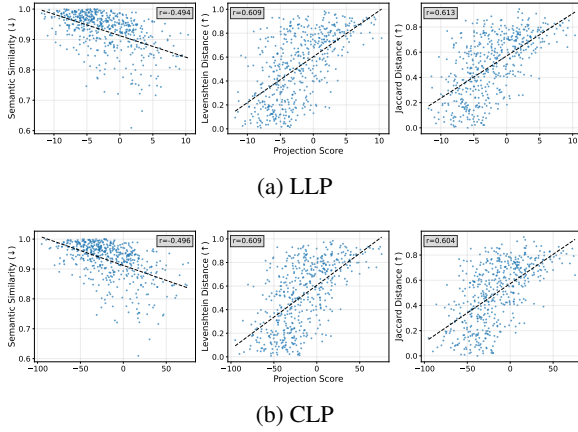


Figure 14: Edit-strength metrics plotted against probe projection scores. The dashed line shows the linear regression fit, and r denotes the Pearson correlation coefficient. This Figure shows the results for EditLens (Thai et al., 2026).

In Figure 6, we plot the text-similarity metrics used to quantify the degree of AI editing in APT-Eval (Saha and Feizi, 2025) against the projection scores of LLP and CLP. Figure 14 presents the same analysis for EditLens (Thai et al., 2026). We observe the same overall pattern across both benchmarks, although the performance gap between LLP and CLP is smaller on EditLens.

C.6 Layer Analysis

We conduct an additional analysis of probe performance across transformer layers to better understand where MGT signals emerge and how they evolve throughout the network. Specifically, we

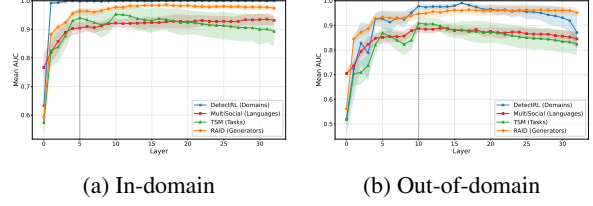


Figure 15: Layer-wise AUC averaged across benchmarks. Vertical lines denote performance plateaus. **Takeaway:** MGT signals emerge early and remain stable.

ask: **RQ:** How does MGT detection performance evolve across layers in in-domain and OOD settings?

MGT signals emerge in early layers and remain relatively stable throughout later layers. Figure 15 shows layer-wise mean AUC across the four subsets within each benchmark for both ID and OOD settings. Both settings reveal a consistent two-stage pattern across benchmarks: a sharp rise in MGT signals in the early layers (up to approximately layer 5 for ID and layer 10 for OOD), followed by a plateau in which detection performance remains relatively stable. In both settings, this suggests that even early representations associated with lower-level textual features (Tenney et al., 2019) encode strong linear MGT signals. For OOD detection, the strong performance of middle layers and the degradation in the final layers are consistent with later representations becoming increasingly specialized and therefore less generalizable (Skean et al., 2025).

C.7 Sensitivity to Text Length

Model ↓	n chars →										
	25	50	75	100	125	150	175	200	250	300	400
LLP	0.850	0.823	0.881	0.908	0.966	0.981	0.989	0.993	0.991	0.995	0.997
CLP	0.840	0.824	0.903	0.960	0.982	0.990	0.996	0.996	0.997	0.997	1.000
RoBERTa	0.874	0.883	0.904	0.969	0.926	0.936	0.998	0.999	1.000	0.999	1.000

Table 4: Ablation study of text length. We compare LLP and CLP against a RoBERTa baseline. Each model is trained on the number of characters indicated by the corresponding column, using the ArXiv subset of DetectRL (Wu et al., 2024).

Table 4 presents AUC detection performance as a function of text length. We compare both MGT probes with the RoBERTa (Liu et al., 2019) baseline, training each model on the ArXiv subset of DetectRL (Wu et al., 2024). While RoBERTa has a slight AUC advantage for texts up to 50 characters, both probes remain competitive and reach near-